VerusCite V1 Benchmark: Citation Verification Accuracy

Andrew Wheeler

2026-07-20

Table of contents

1 The Problem

Fabricated citations in academic papers have grown sharply since the widespread adoption of generative AI writing tools. A Lancet audit of 2.5 million biomedical papers found hallucinated reference rates rising 12-fold between 2023 and early 2026 (Topaz et al. 2026). GPTZero's audit of NeurIPS 2025 found over 100 confirmed hallucinated citations across 53 accepted papers despite rigorous peer review (GPTZero 2026). Separate estimates project roughly 147,000 hallucinated citations across arXiv, bioRxiv, SSRN, and PMC in 2025 alone (Zhao et al. 2026).

Existing peer review and editorial safeguards largely fail to catch these errors before publication. The problem spans disciplines: biomedical research, AI/ML conferences, legal filings, and government reports.

While AI use is driving a surge in publications (see A. Wheeler 2026 for a practical overview), the same large language models that create this problem can also be used to verify citations at scale. VerusCite (veruscite.com) is a tool to automatically extract and verify citations from uploaded documents. This report documents the accuracy of the V1 verification pipeline against a hand-labeled ground truth corpus.

2 Approach

2.1 Design Philosophy

Even in documents with hallucinated citations, the majority of references are legitimate. A tool with a high false positive rate – flagging real papers as hallucinations – would waste a reviewer's time chasing false alarms rather than finding actual problems. VerusCite prioritizes **precision over recall**: it is better to miss some hallucinations than to incorrectly flag verified citations.

The target false positive rate for hallucination flags is below 1%. Recall for detecting actual hallucinations is around 70-80%. This tradeoff means a human reviewer spends most of their time on citations that genuinely need attention rather than dismissing false reports.

The application is designed for human-in-the-loop review. Each citation's verification status can be manually overridden, and the interface surfaces the reasoning behind each determination so reviewers can make informed decisions quickly.

2.2 Comparison to Similar Tools

This approach is similar to Pangram, which also emphasizes low false positive rates for AI-generated content detection. Their published materials discuss the same tradeoff: when base rates of problematic content are low, even small false positive rates cause most flagged items to be false alarms (Emi 2024).

3 Pipeline Architecture

VerusCite processes documents in two stages:

1. **Citation Extraction** – An LLM reads the document text and identifies individual references, parsing out structured fields (title, authors, year, journal, DOI).
2. **Citation Verification** – Each extracted citation is checked against external sources to determine whether it exists as described.

3.1 Verification Categories

The checker assigns each citation one of four statuses:

Status	Meaning
Verified	Citation confirmed to exist via CrossRef or web search
Minor Error	Citation likely exists but has metadata discrepancies (wrong year, volume, pages)
Hallucination	No evidence the cited work exists as described
Not Found	Unable to confirm or deny – includes bare URL citations and paywalled/very recent works with insufficient search results

3.2 Verification Method

Verification proceeds in two passes:

1. **Static CrossRef lookup** – A title/DOI search against CrossRef metadata. Fast and free. Confirms roughly 60% of legitimate citations without needing web search.
2. **Web search verification** – Citations not resolved by CrossRef are sent to a web-search-enabled LLM (Perplexity or OpenAI web search tools) to locate evidence of the publication.

The primary models used are **Perplexity** (hosting Gemini Flash Lite or Gemini 3 Flash) and **OpenAI GPT-5.4-mini/nano** as fallback. Having multiple providers is important operationally – any single provider experiences periodic outages or rate limits. Results reported here represent the range across these configurations.

Cost is kept low by using smaller models with web search rather than large frontier models. A full corpus check (36 papers, 2,200+ citations) runs between $0.41-$1.20 per paper depending on the model.

4 Ground Truth

The V1 validation corpus contains **2287** hand-labeled citations across **36 documents**. Sources are intentionally heterogeneous:

- Papers with known hallucinations identified by others on social media (Chris Carothers, David Buil-Gil)
- GPTZero's NeurIPS 2025 audit (GPTZero 2026)
- Papers flagged by Reviewer3 (Reviewer3, n.d.)
- A ChatGPT deep-research output where all 12 citations are fabricated
- Clean papers (dissertation excerpts, open-access criminology, PLoS ONE articles)
- MDPI and preprint samples

Of the 2287 citations, 1945 (85.0%) are verified, 148 (6.5%) have minor errors, 140 (6.1%) are hallucinations, and 54 (2.4%) are not found. The corpus skews heavily toward verified citations (as real documents do), which makes false positive rate the critical metric.

5 Extraction Results

Citation extraction uses an LLM to parse the document text and identify individual references. The table below shows extraction accuracy against the ground truth.

Table 2: Extraction accuracy by model and OCR backend

Model	OCR	Real	Correct	Missing	Extra	Rate (%)	Cost (USD)
gemini-3.1-flash-lite	pypdfium2	2287	2284	3	12	99.9	1.16
gpt-5.4-nano	pypdfium2	2287	2282	5	13	99.8	0.88

Extraction is very accurate across models. Gemini 3.1 Flash Lite with pypdfium2 correctly extracts over 99.8% of citations at under $0.03 per paper for the full corpus. The few "extra" rows are typically page footers, footnotes, or duplicate listings that the LLM picks up in addition to the actual bibliography.

I also tested `liteparse` with the markdown export format, which tended to produce slightly more artifacts – it does not distinguish line breaks for individual citations as cleanly as pypdfium2, producing a few more missed or merged entries. Results were largely similar overall.

6 Checker Results

Table 3: Checker accuracy by model/provider

Model	Provider	FP Hallucination	FP Minor Error	Hallucination Recall	Not-Verified Recall	CrossRef Verified	Cost (USD)
google/gemini-3-flash-preview	perplexity	0.1% (1)	1.8% (36)	73.6% (103/140)	73.9% (252/341)	1331	21.81
google/gemini-3.1-flash-lite	perplexity	0.4% (8)	2.5% (49)	70.7% (99/140)	78.0% (266/341)	1331	14.77
gpt-5.4-mini	openai	0.1% (2)	3.6% (71)	61.4% (86/140)	81.8% (279/341)	1331	43.3
gpt-5.4-nano	openai	0.3% (6)	4.3% (83)	57.9% (81/140)	82.1% (280/341)	1331	23.53

Key metrics:

- **FP → Hallucination**: Percentage of actually-verified citations incorrectly flagged as hallucinations. All configurations stay at or below 0.4%.
- **Not-Verified Recall**: Percentage of actual problems (hallucinations + minor errors + not-found) correctly identified. Ranges from 72-82%.
- **Hallucination Recall**: Percentage of actual hallucinations correctly flagged. The best configuration (Gemini 3 Flash via Perplexity) achieves 73.6%.

6.1 Cost Breakdown

Table 4: Cost breakdown per full-corpus checker run (36 papers)

Model	Provider	Token Cost (USD)	Search Cost (USD)	Total (USD)	Per Paper (USD)
google/gemini-3-flash-preview	perplexity	17.13	4.68	21.81	0.61
google/gemini-3.1-flash-lite	perplexity	9.58	5.2	14.77	0.41
gpt-5.4-mini	openai	25.2	18.1	43.3	1.2
gpt-5.4-nano	openai	7.05	16.48	23.53	0.65

Perplexity-hosted models are substantially cheaper than direct OpenAI for equivalent or better accuracy. The Gemini 3.1 Flash Lite configuration provides the best cost/accuracy ratio at ~$0.41 per paper.

6.2 CrossRef as First Pass

Across all checker runs, approximately **1,331** of ~1,946 verified citations are confirmed via static CrossRef lookup alone (no web search needed). This is roughly 68% of verified citations resolved without any LLM cost, which keeps per-citation expenses low and latency down.

7 Privacy

VerusCite uses only zero data retention (ZDR) models from all providers. Uploaded documents are not used for model training by any third party.

Additionally, the pipeline minimizes what is sent to external LLMs. Citation extraction uses text search on the locally-extracted PDF text first – only the citation strings themselves are sent to an LLM for structured parsing. During verification, only the parsed citation metadata (title, authors, year) is sent to web search, not the full document text. The full PDF content never leaves the server.

8 Limitations and Next Steps

The V1 corpus is heterogeneous by design, but 36 documents is a limited sample. The next evaluation (V2) will use the same prompts and pipeline but a **completely different validation set** – different documents, different domains, different sources of known hallucinations. This ensures the system is not overtrained on the particular examples in V1.

Known limitations of the current evaluation:

- Ground truth labels for some documents were seeded from the checker itself and hand-reviewed, which may introduce subtle bias toward the checker's own patterns.
- "Not found" is an inherently ambiguous category – some citations are bare URLs, others exist but are difficult to locate via web search (paywalled, very recent, or in non-English databases).
- The corpus over-represents arXiv preprints relative to other domains (law, humanities, clinical medicine).

9 Reproducibility

All data for this report is public at github.com/apwheele/veruscite-data. The `v1/` directory contains:

- `ground_truth.csv` – hand-labeled citations with `expected_status`
- `extraction_run/` – raw extraction outputs per model
- `checking_run/` – raw checker outputs per model
- `manifest.json` – run IDs used in this report

To regenerate this report:

```
pip install -r requirements.txt
quarto render report.qmd
```

10 AI Use Disclosure

This paper was generated entirely using Claude Opus 4.6, reviewing prior works by Andrew Wheeler. See A. P. Wheeler (2026) for an overview of this workflow.

References

Emi, Bradley. 2024. "All about False Positives in AI Detectors." https://www.pangram.com/blog/all-about-false-positives-in-ai-detectors.

GPTZero. 2026. "GPTZero Finds 100 New Hallucinations in NeurIPS 2025 Accepted Papers." https://gptzero.me/news/neurips/.

Reviewer3. n.d. "Reviewer3: Live arXiv Reference Checking." https://reviewer3.com/live/arxiv.

Topaz, Maxim et al. 2026. "Fabricated Citations: An Audit Across 2.5 Million Biomedical Papers." *The Lancet*. https://doi.org/10.1016/s0140-6736(26)00603-3.

Wheeler, Andrew. 2026. *Large Language Models for Mortals: A Practical Guide for Analysts with Python*. Crime De-Coder.

Wheeler, Andrew P. 2026. "Using Claude Code to Help Me Write." https://andrewpwheeler.com/2026/03/20/using-claude-code-to-help-me-write/.

Zhao, Zhenyue, Yihe Wang, Toby Stuart, Mathijs De Vaan, Paul Ginsparg, and Yian Yin. 2026. "LLM Hallucinations in the Wild: Large-Scale Evidence from Non-Existent Citations." https://arxiv.org/abs/2605.07723.